Experiment-4

Create a web form that accepts a user's name and age. Write a Servlet to process the form data and display it back on the browser.

AIM:

To create a simple web application that includes a form to accept a user's name and age, and then use a Servlet to process the submitted data. The Servlet will retrieve the form data, process it, and display the values back on the browser.

ALGORITHM:

1. Input:

- The user fills out a web form with their **name** and **age** and submits it.

2. Form Handling and Servlet Processing:

1. **Create the Web Form:**
    - Create an HTML form with **two input fields**: one for the user's **name** and one for the user's **age**.
    - Add a **submit button** to the form.
    - Set the form's action attribute to the URL of the **Servlet** and the method to POST (or GET).
2. **Create the Servlet Class:**
    - Write a Servlet that processes the **form data** submitted by the user.
    - The Servlet will override the doPost() (or doGet() depending on the form method) to handle form submissions.
    - The Servlet will **retrieve the name and age** parameters from the request object using request.getParameter("name") and request.getParameter("age").
3. **Process the Data in the Servlet:**
    - Extract the name and age values submitted from the form.
    - **Validate the input** (e.g., check if the age is a valid number).
    - Prepare a **response** that will display the user's **name** and **age** back on the browser.
4. **Send Response to the Browser:**

- The Servlet will send a **response** to the browser containing a message like:
  - "Hello, [name]! You are [age] years old."
- The response will be displayed to the user in the browser.

5. **Deploy the Servlet:**
   - Compile the Servlet and deploy it to a **Servlet container** (like **Apache Tomcat**).
   - Ensure the web.xml file or Servlet annotations are used to map the Servlet to the URL defined in the form's action attribute.
6. **Test the Application:**
   - Open the web browser, fill in the form with the name and age, and submit it.
   - The Servlet will process the data and display the message back to the user in the browser.

PROGRAM:

hello.html:

```
<head>
<title>Use Form</title>
<style>
body{
        font-family: Arial;
        background-color: #f2f2f2;
}
.container{
    width: 400px;
    margin: 100px auto;
    padding: 20px;
    background-color: white;
    border-radius: 10px;
    box-shadow: 0px 0px 10px gray;
    text-align:center;
}
h2{
       color: #333;
}
.form-group{
       display: flex;
       align-items: center;
       margin-bottom: 12px;
}
label{
       width:80px;
}
input[type="text"],input[type="number"]{
       flex:1;
```

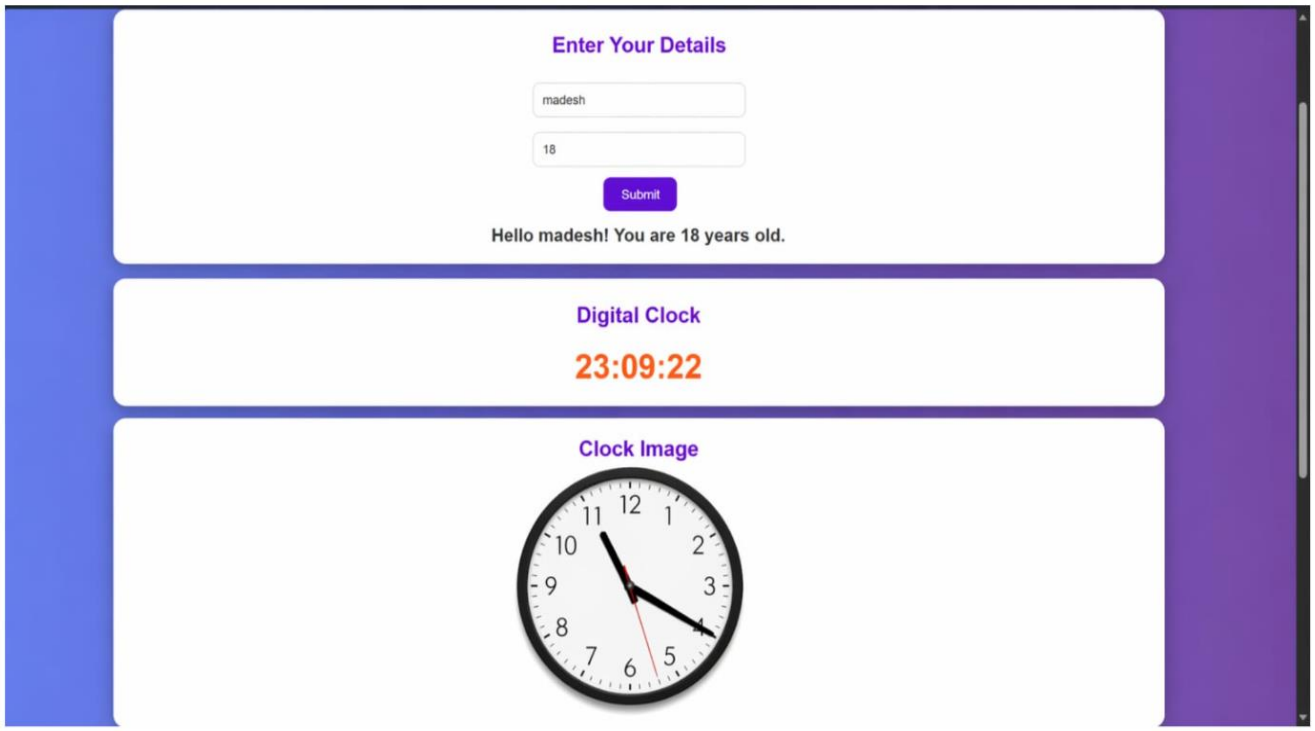
```
        padding:6px;
        border: 1px solid #ccc;
        border-radius: 4px;
}
.submit-btn{
        text-align: center;
}
input[type="submit"]{
        background-color: #4CAF50;
        color: white;
        border: none;
        padding: 8px 15px;
        border-radius: 4px;
        cursor: pointer;
}
input[type=submit]:hover {
        background-color: #45a049;
}
</style>
</head>
<body>
<div class="container">
<h2>User Details Form</h2>
<form action="UserServlet" method="post">
<div class="form-group">
<label>Name:</label>
<input type="text" name="username" required>
</div>
<div class="form-group">
<label>Age: </label>
<input type="number" name="age" required>
</div>
<div class="submit-btn">
<input type="submit" value="Submit">
</div>
</form>
</div>
</body>
```

UserServlet.java:

```
package WTMA;

import  jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import  jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;
import java.io.PrintWriter;

@WebServlet("/UserServlet")
public class UserServlet extends HttpServlet {
        private static final long serialVersionUID = 1L;
```

```
        protected void doPost(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
                // TODO Auto-generated method stub
                response.setContentType("text/html");
                PrintWriter pw=response.getWriter();
                String name =request.getParameter("username");
                String age=request.getParameter("age");
                pw.println("Hello "+name+" ,You are "+age+" years old");

        }

}
```

OUTPUT-1:



RESULT:

A Servlet is created to process the HTML form data and display it back on the browser.